Sadie Lee

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Experience

Boston Children's Hospital **Jun 2026 – Present**

Research Assistant – Computational Health Informatics Program

- Evaluating frontier LLMs for clinical decision support using clinician-annotated benchmark datasets to measure model safety and deployment readiness.
- Building scalable end-to-end data analytic pipelines in Python and R to process and analyze large Electronic Health Record (EHR) data.
- Engineered a deterministic NLP pipeline to parse unstructured clinical text, enabling targeted cohort segmentation.
- Conducting additional original research using state space models to forecast long-term continuous glucose trajectories and dynamics.

Mayo Clinic **May 2025 – Aug 2025**

Research Student – Mayo Clinic Platform

- Analyzed patient re-identification risks in vision transformers and convolutional neural networks to assess patient privacy and data leakage prior to exporting model weights outside secure environments.
- Executed model inversion attacks using a generative adversarial network to reconstruct medical images and predict demographic attributes from frozen model weights.

Voythos **May 2025 – Jul 2025**

Data Science Intern

- Developed multi-task vision transformers on volumetric chest CT images for aortic segmentation and thrombosis detection, feeding downstream multimodal models to predict risk of Type B aortic dissection.
- Refactored machine learning pipelines and resolved data processing bugs, accelerating model training time by 97%.

Mayo Clinic **May 2024 – Aug 2024**

Undergraduate Intern – Mayo Clinic Platform

- Built dashboards in Power BI to geographically track customer acquisitions, visualize account metrics, and present product KPIs.
- Established an internal pipeline to help health tech start-up customers prepare for market by providing de-identified medical images for AI/ML development.
- Trained 3D convolutional neural networks in PyTorch for abnormality classification and segmentation.
- Queried relational clinical databases using SQL to assess data availability and scope feasibility for prospective customers.

BC Children's Hospital Research Institute, UBC Faculty of Medicine **May 2023 – Apr 2024**

Research Assistant

- Evaluated the impact of a digital peer support intervention on glycemic control in adults with type 1 diabetes as part of an interventional clinical trial.
- Coded R scripts to collect and process glucose sensor time-series data, calculating changes in sensor-derived metrics pre- and post-intervention.
- Served as the primary liaison between software engineering, research, and clinical teams to manage app deployment, document bug fixes, and lead user onboarding.
- Co-authored study protocols, Data Safety and Monitoring Board (DSMB) plans, and ethics amendments.

Education

University of British Columbia, Vancouver, Canada **May 2026**
Bachelor of Arts in Cognitive Systems, Minor in Data Science – High Distinction

Relevant coursework: Statistical Modeling, Databases, Applied Machine Learning, Data Structures and Algorithms, Data Visualization, Discrete Math

Honors and Awards

Andrew Wade Memorial Award in Visual Analytics Mar 2026
UBC Arts Undergraduate Society Student Conference Grant Award Mar 2025

Technical Skills

Languages: Python, R, SQL, JavaScript/TypeScript, HTML/CSS
Tools/Frameworks: Git, Docker, Linux/Bash, PyTorch, pymc, MongoDB, Power BI, React, REDCap

Research

T1D REACHOUT–A Mobile App to Deliver Peer-Led Mental Health Support to Adults Living With Type 1 Diabetes: Co-Design and Development Process
Sadie Lee, Baray Sidhu, Parteek Johal, Ayman Azhar, Jonath Sujan, Matthias Görges, Tricia S Tang
JMIR Diabetes. Mar 2026.

Re-identification Risk of Medical Imaging-Based Deep Learning Models
Sadie Lee; Supervised by Adam Resnick and Rob Blundo (Mayo Clinic Platform)
Undergraduate Capstone. Aug 2025.

Formalizing Ethical Design in Prostate Cancer Image Analysis: A Preliminary Case Study
Sadie Lee, Adam Resnick, Nasibeh Zanjirani Farahani
IEEE MIT Undergraduate Research Technology Conference. Oct 2024.

Topological Data Analysis and Interpretability of 3D-Convolutional Neural Networks
Sadie Lee
AAAI Undergradute Consortium. Feb 2024.

Teaching and Volunteering

University of British Columbia **Jan – Apr 2026**
Teaching Assistant - COGS 401 (Seminar in Cognitive Systems)

UBC Multidisciplinary Undergraduate Research Conference **Mar 2026**
Adjudicator

UBC Emerging Media Lab **May – Aug 2023**
Software Developer

UBC Multifaceted Innovations in Neurotechnology **Mar – Dec 2023**
Deep Learning Researcher

UBC 180 Degrees Consulting **Jan – Apr 2023**
Data Science Consultant